INCH-POUND

MIL-E-1/1099E
3 March 1995
SUPERSEDING
MIL-E-1/1099D
6 August 1975

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON
TYPE 5948A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen
See figures 1, 1A, 2, and 3
Mounting position: Vertical, base down
Weight: 4.6 pounds (2.1 kg) nominal

ABSOLUTE RATINGS:

Parameter:	Ef Vac	epy kv	epx kv	Ebb kV dc	Ip Aac	egy v	egx v	Ecc Vdc	Ib a
Maximum:	6.6	25.0 1/	25.0 2/	---	30	---	650	---	1,000
Minimum:	6.0	10.0	5% epy	5.0	---	3/	---	0	---
Test conditions:	6.3	25.0	---	---	---	550	---	0	---
	Ib Adc	prr	tj μs	pb	tk	TA °C	dik/dt a/μs	Eres Vac	Ires Aac
Maximum:	1.0	---	0.005 4/	9.0×10^9	---	+75	5,000	5.5 5/	8.0
Minimum:	---	---	---	---	180	-55	---	2.5	---
Test conditions:	---	360	---	---	180	---	---	6/	---

1/ Instantaneous starting is not recommended. When it is absolutely necessary, however, the maximum permissible epy is 18.0 kv and shall not be attained in less than 0.04 seconds.
2/ The peak inverse voltage, exclusive of a spike of 0.05 μs maximum duration, shall not exceed 5 kv during the first 25 μs following the anode pulse.
3/ The driver pulse, measured at the tube socket with thyratron grid disconnected, shall be: egy = 700 volts minimum, 1,000 volts maximum; tr = 0.35 μs maximum; tp = 2.0 μs maximum; impedance of driver circuit 50 to 200 ohms.
4/ Rated time jitter of 0.005 μs applies at a peak forward voltage of 15.0 kv or greater. At peak forward anode voltages between 10 and 15 kv, the tube is rated at a time jitter of 0.01 μs.
5/ The optimum reservoir voltage for operation in accordance with operation (1) conditions is inscribed on the base of the tube and must be held to within ±5 percent.
6/ Adjust reservoir voltage to value indicated on tube to within ±5 percent.

GENERAL: Qualification - Required

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELD, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5960

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TABLE I. Group A inspection.

Inspection	Method	Conditions	Acceptance level	Symbol	Limits		Unit
					min	max	
Quality conformance inspection, part 1							
Heater current	3241		0.65	If	14.0	33.0	Aac
Heater current	3241	Eres = 4.5 Vac	0.65	Ires	3.0	6.0	Aac
DC anode voltage for conduction	3241	Eres = inscribed value -5 percent	0.65	Ebb	----	4000	Vdc
Instantaneous starting 1/ 4/	3267	epy = 180 kv(min); Eres = inscribed value +5 percent	0.65	----	----	----	----
Operation (1) 1/	3246	epy = 27.5 kv; Eres = inscribed value -5 percent; t = 30 minutes	0.65	egy	----	550	v
Operation (2) 5/ 6/	3246	epy = 15 kv; prr = 1,500; Eres = 95 percent of optimum value	0.65	egy	----	550	v
Operation of optimum reservoir voltage (1)	3246	Operation (1)	0.65	Eres	2.5	5.5	Vac
Operation of optimum 6/ reservoir voltage (2)	3246	Operation (2)	0.65	Eres	2.5	5.5	Vac
Pulse emission	3251	ik = 1,500 a; tp = 5.0 µs ±10 percent; prr = 60 ±10 percent; tr = 0.5 µs (max); Eres = inscribed value; starting time of current pulse = 2.5 µs	0.65	egk	----	400	v
Quality conformance inspection, part 2							
Operation (3)	3246	Operation (1), except Eres = inscribed value +5 percent; tk = 900 sec; t = 1,2000 sec	----	egy	----	550	v
Operation (4)	3246	Operation (2), except Eres = 105 percent; optimum value	----	egy	----	550	v
Anode delay time	3256	Operation (1); t = 120 sec	----	tad	----	1.0	µs
Anode delay time drift 7/	3256	Anode delay time	----	Δtad	----	0.25	µs
Time jitter 8/	3261	Operation (1); except epy = 12.5 kv	----	tj	----	0.004	µs
Quality conformance inspection, part 3							
Life test (1) 1/	----	Group C; t = 96 hours "on" and 1 hour "off"; t = 500 hours	----	----	----	----	----

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TABLE I. Group A inspection.

Inspection	Method	Conditions	Acceptance level	Symbol	Limits		Unit
					min	max	
Quality conformance inspection, part 3 -Continued							
Life test (2) 6/	----	Life test (1); operation (2) Zg = 250 Ω; t = 500 hours	----	----	----	----	----
Life test (1) and (2) endpoints:							
Operation (1)	3246		----	egy	----	650	v
Dc anode voltage for conduction	3247	egy = 500 v	----	Ebb	----	4500	Vdc
Time Jitter	3261	egy = 500 v; Zg = 250 Ω	----	tj	----	0.005	μs
Operation at elevated ambient temperature 1/ 2/	3246	TA = 75°C; t = 5 hours	----	egy	----	550	v
Life test (3) 1/ 3/	----	t = 500 hours	----	----	----	----	----
Life test (3) endpoints:							
Operation (1)	3246		----	egy	----	675	v
Dc anode voltage for conduction	3247		----	Ebb	----	4500	Vdc
Time Jitter	3261	egy = 650 v	----	tj	----	0.01	μs

1/ The circuit constants shall be chosen so that at epy = 25.0 kv under charging conditions: dik/dt = 5,000 a/μs minimum; ib = 1,000 a minimum; tp = 2.5 ±0.25 μs; prr = 500 minimum. The grid pulse characteristics shall be: tp = 2.0 μ maximum; tr = 0.35 μs minimum; internal impedance of driver = 250 ohms minimum.

2/ This test shall be conducted for a total of 5 consecutive hours with no more than three kickouts and with no evidence of anode heating. The tube shall be started at Eres = 105 percent of the inscribed value and operate at this value for 4 hours. At the start of the fifth hour and while the tube is still operating, the reservoir voltage shall be lowered to 95 percent of the inscribed value and remain there for the final 1 hour of operation.

3/ The life test shall be intermittent, with the tube operating approximately half the total elapsed time, and held without reservoir voltage or cathode-heater voltage approximately half the total time. The tube shall make a minimum of four starts during each 24-hour period. The life-test duration is measured in operating hours. Failure of the tube to operate following the minimum allowable cathode and reservoir heater warmup time constitutes failure of the test.

4/ The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 18 kv within 0.03 second. (The filter in the rectifier shall be designed so that the epy reaches at least 9 kv within 0.015 second.) Any tube failing to start within three attempts will be considered a failure.

5/ The anode circuit constants shall be so chosen that a resonant charging conditions; epy = 15 kv; ib = 500 a minimum; dik/dt = 5,000 a/μs minimum. Output pulse; tp = 1.3 μs ±10 percent; prr = 1,500 minimum; epx = 5.0 kv. Grid pulse the same as note 1. Adjust reservoir voltage to optimum value for these test conditions. The tube shall operate continuously for 20 minutes without continuous conduction.

6/ The optimum reservoir voltage for operation in accordance with operation (2) conditions shall be so determined that the tube shall operate at 95 to 105 percent of the established reservoir voltage for these conditions.

7/ This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 minutes, 4 minutes, and 30 minutes of the operation (1) test. The Δtad measurement is the numerical difference between the 2-minute and 4-minute, or 2-minute and 30-minute anode delay time readings, whichever is greater.

8/ The tj shall be not greater than the amount specified after 120 seconds of anode operation.

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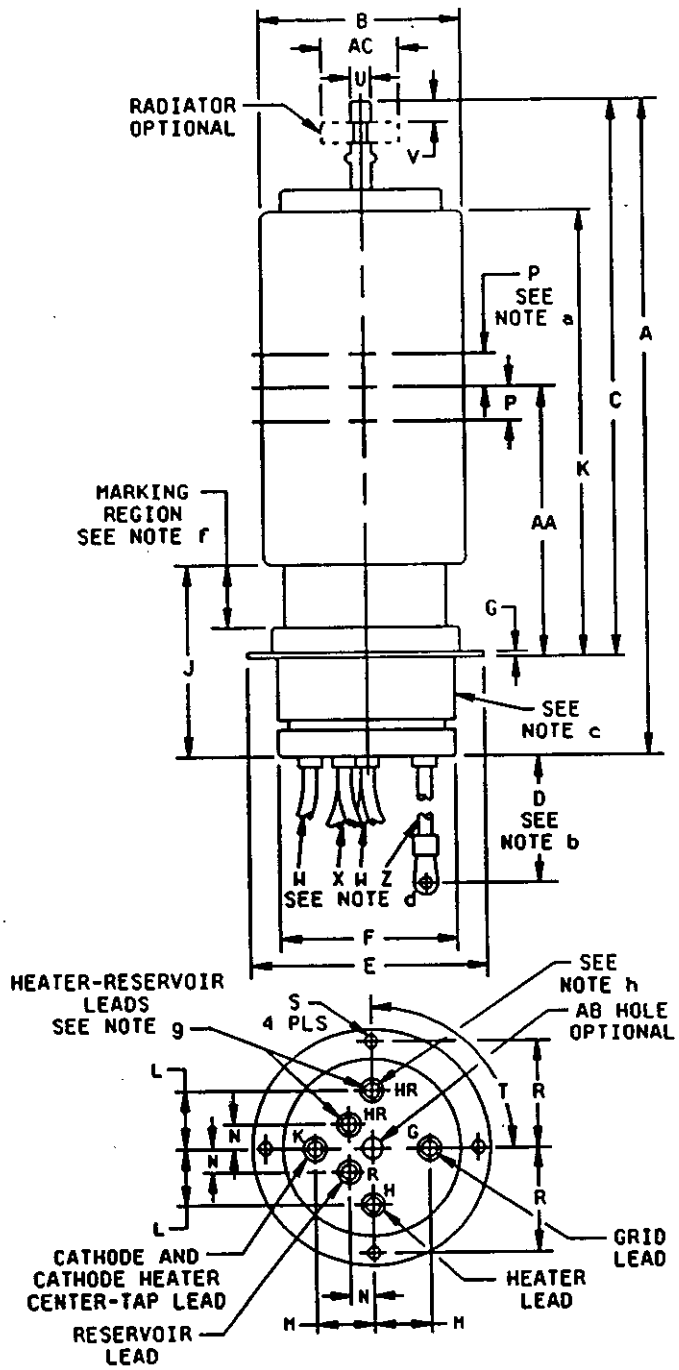


FIGURE 1. Outline drawing of electron tube type 5948A.

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LTR	Dimensions			
	Millimeters		Inches	
Quality conformance inspection, part 2				
	Min	Max	Min	Max
A	387.35	412.75	15.250	16.250
B	123.83	130.18	4.875	5.125
C	333.38	352.42	13.125	13.875
F	112.73	115.87	4.438	4.562
R	67.61	68.12	2.662	2.682
T	89.6°		90.4°	
Quality conformance inspection, part 3				
See note K				
D	127.00	152.40	5.000	6.000
E	150.83	153.97	5.938	6.062
J	110.34	118.26	4.344	4.656
K	252.43	280.97	9.938	11.062
L	36.53	39.67	1.438	1.562
M	33.32	36.53	1.312	1.438
N	11.13	14.27	0.438	0.562
S	7.85	8.00	0.309	0.315
U	14.20	14.55	0.559	0.573
V	12.70		0.500	
AC	-----	50.80	-----	2.000
P	-----	25.40	-----	1.000
AA	-----	175.10	-----	6.500
Reference dimensions				
G	2.39		0.094	
AB	12.70		0.500	
W	4,500 (circular mils)			
X	1,500 (circular mils)			
Z	2,000 (circular mils)			

NOTES:

a. Dimension P designates the area to be used for clamping.
b. Dimension D is measured from bottom of base insert to center of lead lug hole.
c. Base, metal shell, with mounting flange and phenolic insert. Flange secured to base with rivets, or equivalent. Insert shall be low-loss phenolic wafer or laminated phenolic resin, permanently marked as follows: G (grid), H (heater), R (reservoir), HR (heater-reservoir), and K (cathode). All feed-through fittings shall be equipped with setscrews, or equivalent.
d. Leads shall be flexible. Dimensions W, X, and Z apply to cross-sectional area. Lead identification insulation sleeving and lugs shall be as follows:

Lead	Color	Sleeving	Lugs (see note j)
G (grid)	Green	H-A-1(see note e)	MS25036-108
H (heater)	Yellow	H-A-1 or H-B-1	MS25036-157
HR (heater) (see note g)	Yellow with black sleeve	H-A-1 or H-B-1	MS25036-157
HR (reservoir) (see note g)	Red with yellow sleeve	H-A-1 or H-B-1	MS25036-108
R (reservoir)	Red	H-A-1 or H-B-1	MS25036-108
K (cathode)	Black	H-A-1 or H-B-1	MS25036-157

e. As specified in specification MIL-I-3190.
f. Reservoir voltage shall be inscribed once above each mounting hole on base wall.
g. Heater lead internally connected to reservoir lead.
h. Cathode heater connection to this lead.
j. Lugs shall be permanently identified with symbols as follows: G (grid), K (cathode), H (heater), R (reservoir), and HR (heater-reservoir).
k. Holes S shall be inspected for orientation by using the base gage outlined in figure 3.

FIGURE 1. Outline drawing of electron tube type 5948A - Continued.

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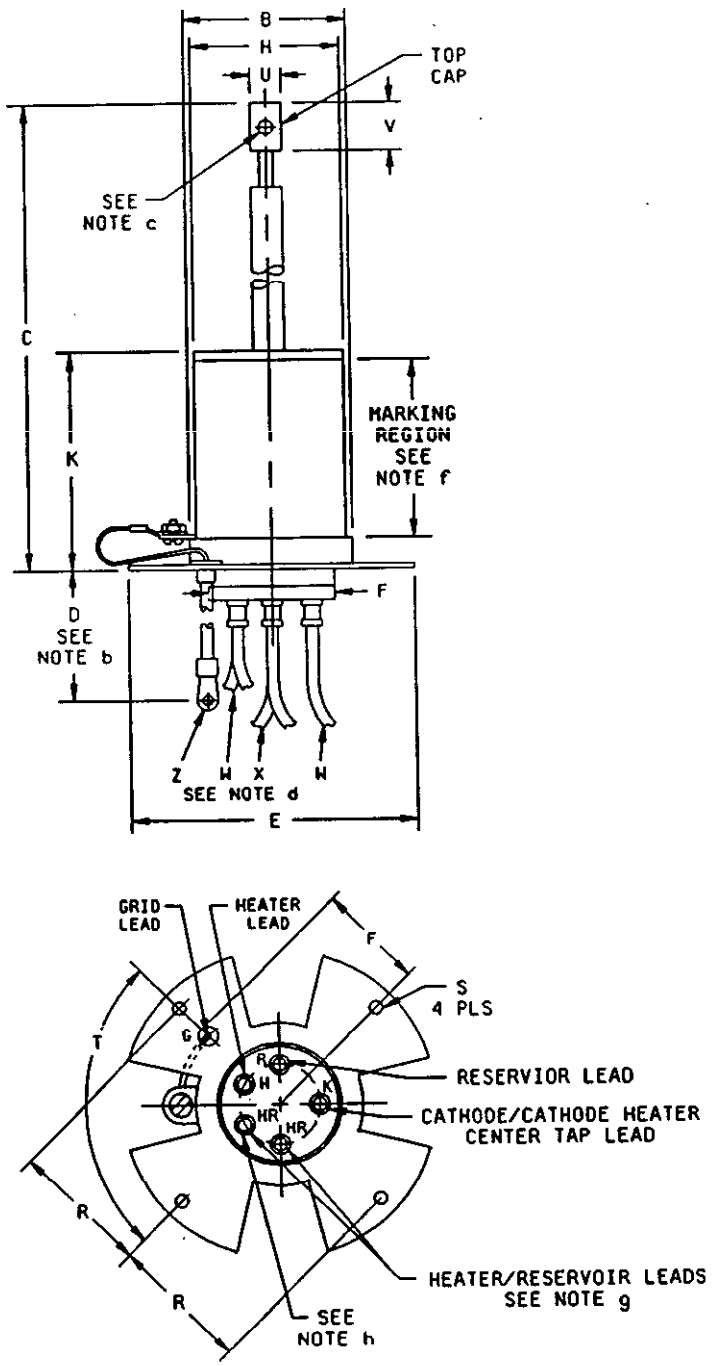


FIGURE 1A. Alternate outline drawing of electron tube type 5948A.

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LTR	Dimensions			
	Millimeters		Inches	
Quality conformance inspection, part 2				
	Min	Max	Min	Max
B	79.37	85.73	3.125	3.375
C	318.31	337.36	12.532	13.282
F	------	57.94	-----	2.281
R	67.61	68.12	2.662	2.682
T	-----	-----	89.6°	90.4°
Quality conformance inspection, part 3				
See note K				
D	203.20	228.60	8.000	9.000
E	150.83	153.97	5.938	6.062
H	73.03	79.38	2.875	3.125
K	99.70	118.75	3.925	4.675
S	7.85	8.00	0.309	0.315
U	14.02	14.53	0.552	0.572
V	15.49	16.00	0.610	0.630
Reference dimensions				
W	4,500 (circular mils)			
X	1,500 (circular mils)			
Z	2,000 (circular mils)			

NOTES:

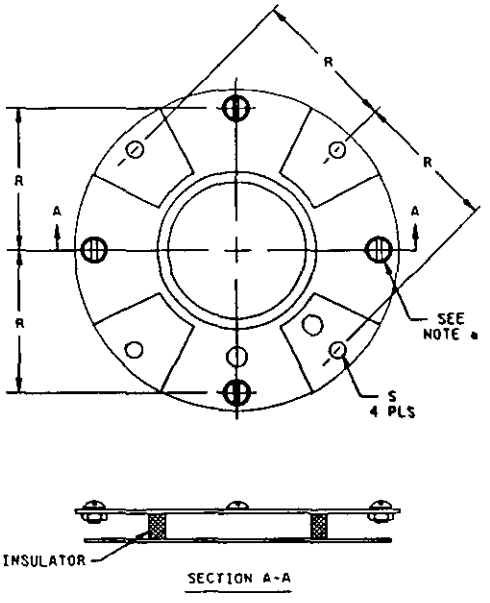
a. A socket adapter (figure 3) is supplied with each thyratron for applications that required electrical isolation of the thyratron mounting flange from the cathode (K) lead.
b. Dimension D is measured from bottom of flange to center of lead lug hole.
c. Set screw 6-32 NC X 3/16 steel, cup point.
d. Leads shall be flexible. Leads shall be identified as follows:

Lead	Color	Sleeving	Lugs (see note j)
G (grid)	Green	H-A-1(see note e)	MS25036-108
H (heater)	Yellow	H-A-1 or H-B-1	MS25036-157
HR (heater) (see note g)	Yellow with black sleeve	H-A-1 or H-B-1	MS25036-157
HR (reservoir) (see note g)	Red with yellow sleeve	H-A-1 or H-B-1	MS25036-108
R (reservoir)	Red	H-A-1 or H-B-1	MS25036-108
K (cathode)	Black	H-A-1 or H-B-1	MS25036-157

e. As specified in specification MIL-I-3190.
f. Reservoir voltage shall be inscribed once above each mounting hole on base wall.
g. Heater lead internally connected to reservoir lead.
h. Cathode heater connection to this lead.
j. Lugs shall be permanently identified with impression stamped symbols as follows: G (grid), K (cathode), H (heater), R (reservoir), and HR (heater-reservoir).
k. Holes S shall be inspected for orientation by using the base gage outlined in figure 3.

FIGURE 1A. Alternate outline drawing of electron tube type 5948A - Continued.

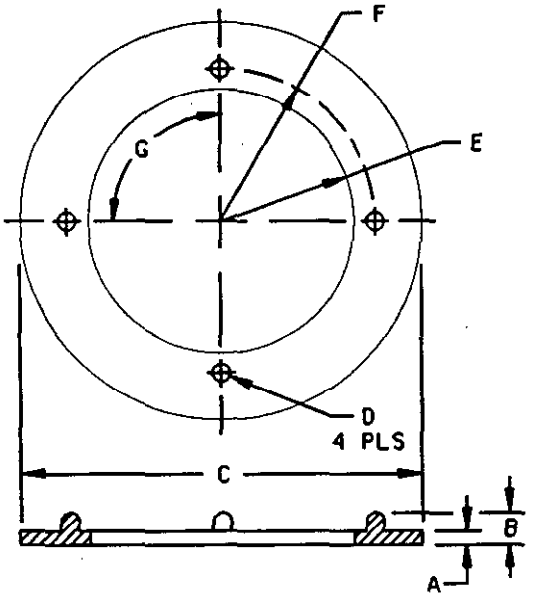
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Ltr	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
R	67.61	68.12	2.662	2.682
S	7.85	8.00	.309	.315

NOTE:
a. 10-32 X 1/4 truss head machine screws.

FIGURE 2. Socket adapter.



Ltr	Dimensions			
	Millimeters		Inches	
	Min	Max	Min	Max
A	6.05	6.65	.238	.262
B	7.52	8.33	.296	.328
C	168.91	186.69	6.650	7.350
D	7.02 DIA	7.05 DIA	0.2765 DIA	0.2775 DIA
E	58.72 R	58.76 R	2.312 R	2.314 R
F	67.84 R	67.89 R	2.671 R	2.673 R
G	------	------	89.08°	90.02°

FIGURE 3. Base gage.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - AS, CG, MC, OS
Air Force - 11, 17, 19, 99

Preparing activity:
DLA - ES

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